

TryHackMe - KAPE

Kroll Artifact Parser and Extractor (KAPE) parses and extracts Windows forensics artifacts. It is a tool that can significantly reduce the time needed to respond to an incident by providing forensic artifacts from a live system or a storage device much earlier than the imaging process completes.

KAPE serves two primary purposes, 1) collect files and 2) process the collected files as per the provided options. For achieving these purposes, KAPE uses the concept of targets and modules. Targets can be defined as the forensic artifacts that need to be collected. Modules are programs that process the collected artifacts and extract information from them. We will learn about them in the upcoming tasks.

How it works

KAPE is extensible and highly configurable. In essence, the KAPE binary collects files and processes them as per the provided configuration.

The collection of files (targets) KAPE adds the files to a queue and copies them in two passes. In the first pass, it copies the files that it can. This works for files that the OS has not locked. The rest of the files are passed to a secondary queue. The secondary queue is processed using a different technique that uses raw disk reads to bypass the OS locks and copy the files. The copied files are saved with original timestamps and metadata and stored in a similar directory structure.

Once the data is collected, KAPE can process it using modules. The modules can be independent binaries that run on the collected data and process them to extract information. For example, KAPE will collect and copy the Prefetch file to our target destination during the target collection. Running a Prefetch Parser (PECmd) module on this target will extract the prefetch file and save it in a CSV file.

Target Options

In KAPE's lexicon, **Targets** are the artifacts that need to be collected from a system or image and copied to our provided destination. For example, as we learned in the last room, Windows Prefetch is a forensic artifact for evidence of execution so that we can create a **Target** for it. Similarly, we can also create **Targets** for the registry hives. In short, **Targets** copy files from one place to another.

A TKAPE file contains information about the artifact that we want to collect, such as the path, category, and file masks to collect

Compound Targets:

KAPE also supports `Compound Targets`. These are `Targets` that are compounds of multiple other targets. KAPE is often used for quick triage collection and analysis. The purpose of KAPE will not be fulfilled if we have to collect each artifact individually. Therefore, `Compound Targets` help us collect multiple targets by giving a single command.

!Disabled

This directory contains `Targets` that you want to keep in the KAPE instance, but you don't want them to appear in the active `Targets` list.

!Local

If you have created some `Targets` that you don't want to sync with the KAPE Github repository, you can place them in this directory. These can be `Targets` that are specific to your environment. Similarly, anything not present in the Github repository when we update KAPE will be moved to the `!Local` directory.

Module Options

`Modules`, in KAPE's lexicon, run specific tools against the provided set of files. Their goal is not to copy files from one place to another but rather run some command and store the output. Generally, the output is in the form of CSV or TXT files.